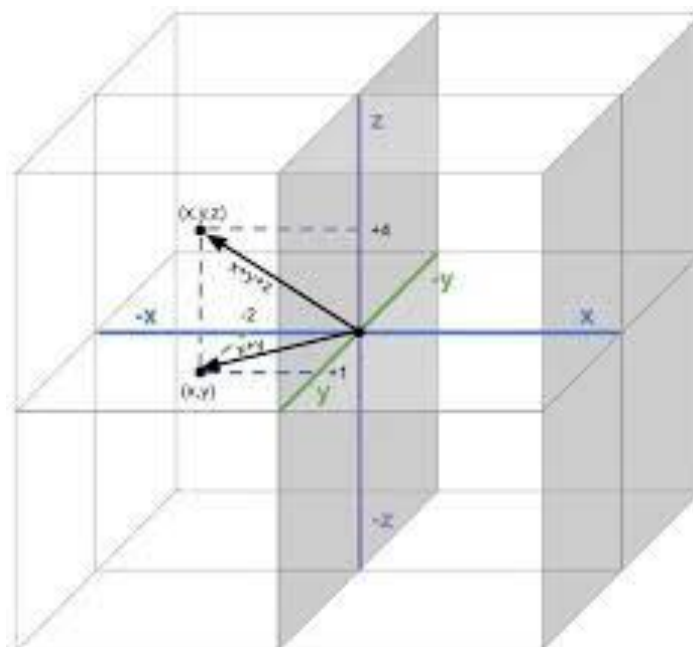


Linear Algebra

Math 1057

Instructor

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Lecture 8

Onto and One-to-One Mappings, and the Matrix of a Linear Mapping

8.1 Onto Mappings

We have seen through examples that the range of a vector mapping (linear or nonlinear) is not always the entire co-domain. For example, if $T_A(\mathbf{x}) = A\mathbf{x}$ is a matrix mapping and $\mathbf{b}$ is such that the equation $A\mathbf{x} = \mathbf{b}$ has no solutions then the range of T does not contain $\mathbf{b}$ and thus the range is not the whole co-domain.

Definition 8.1: A vector mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **onto** if for each $\mathbf{b} \in \mathbb{R}^m$ there is at least one $\mathbf{x} \in \mathbb{R}^n$ such that $T(\mathbf{x}) = \mathbf{b}$.

For a matrix mapping $T_A(\mathbf{x}) = A\mathbf{x}$, the range of T_A is the span of the columns of A . Therefore:

Theorem 8.2: Let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the matrix mapping $T_A(\mathbf{x}) = A\mathbf{x}$, where $A \in M_{m \times n}$. Then T_A is onto if and only if the columns of A span all of $\mathbb{R}^m$.

Combining Theorem 4.11 and Theorem 8.2 we have:

Theorem 8.3: Let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the matrix mapping $T_A(\mathbf{x}) = A\mathbf{x}$, where $A \in \mathbb{R}^{m \times n}$. Then T_A is onto if and only if $r = \text{rank}(A) = m$.

Example 8.4. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the matrix mapping with corresponding matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ -3 & -4 & 2 \\ 5 & 2 & 3 \end{bmatrix}$$

Is T_A onto?

Solution. The $\text{rref}(\mathbf{A})$ is

$$\begin{bmatrix} 1 & 2 & -1 \\ -3 & -4 & 2 \\ 5 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore, $r = \text{rank}(\mathbf{A}) = 3$. The dimension of the co-domain is $m = 3$ and therefore $\mathsf{T}_{\mathbf{A}}$ is onto. Therefore, the columns of $\mathbf{A}$ span all of $\mathbb{R}^3$, that is, every $\mathbf{b} \in \mathbb{R}^3$ can be written as a linear combination of the columns of $\mathbf{A}$:

$$\text{span} \left\{ \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} \right\} = \mathbb{R}^3$$

□

Example 8.5. Let $\mathsf{T}_{\mathbf{A}} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be the matrix mapping with corresponding matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 & 4 \\ -1 & 4 & 1 & 8 \\ 2 & 0 & -2 & 0 \end{bmatrix}$$

Is $\mathsf{T}_{\mathbf{A}}$ onto?

Solution. The $\text{rref}(\mathbf{A})$ is

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 & 4 \\ -1 & 4 & 1 & 8 \\ 2 & 0 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore, $r = \text{rank}(\mathbf{A}) = 2$. The dimension of the co-domain is $m = 3$ and therefore $\mathsf{T}_{\mathbf{A}}$ is **not** onto. Notice that $\mathbf{v}_3 = -\mathbf{v}_1$ and $\mathbf{v}_4 = 2\mathbf{v}_2$. Thus, $\mathbf{v}_3$ and $\mathbf{v}_4$ are already in the span of the columns $\mathbf{v}_1, \mathbf{v}_2$. Therefore,

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\} \neq \mathbb{R}^3.$$

□

Below is a theorem which places restrictions on the size of the domain of an onto mapping.

Theorem 8.6: Suppose that $\mathsf{T}_{\mathbf{A}} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a matrix mapping corresponding to $\mathbf{A} \in M_{m \times n}$. If $\mathsf{T}_{\mathbf{A}}$ is onto then $m \leq n$.

Proof. If $\mathsf{T}_{\mathbf{A}}$ is onto then the $\text{rref}(\mathbf{A})$ has $r = m$ leading 1's. Therefore, $\mathbf{A}$ has at least m columns. The number of columns of $\mathbf{A}$ is n . Therefore, $m \leq n$. □

An equivalent way of stating Theorem 8.6 is the following.

Corollary 8.7: If $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a matrix mapping corresponding to $A \in M_{m \times n}$ and $n < m$ then T_A cannot be onto.

Intuitively, if the domain $\mathbb{R}^n$ is “smaller” than the co-domain $\mathbb{R}^m$ and $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear then T_A cannot be onto. For example, a matrix mapping $T_A : \mathbb{R} \rightarrow \mathbb{R}^2$ cannot be onto. Linearity plays a key role in this. In fact, there exists a continuous (nonlinear) function $f : \mathbb{R} \rightarrow \mathbb{R}^2$ whose range is a square! In this case, the domain is 1-dimensional and the range is 2-dimensional. This situation cannot happen when the mapping is linear.

Example 8.8. Let $T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the matrix mapping with corresponding matrix

$$A = \begin{bmatrix} 1 & 4 \\ -3 & 2 \\ 2 & 1 \end{bmatrix}$$

Is T_A onto?

Solution. T_A is onto because the domain is $\mathbb{R}^2$ and the co-domain is $\mathbb{R}^3$. Intuitively, two vectors are not enough to span $\mathbb{R}^3$. Geometrically, two vectors in $\mathbb{R}^3$ span a 2D plane going through the origin. The vectors not on the plane $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ are not in the range of T_A . $\square$

8.2 One-to-One Mappings

Given a linear mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the question of whether $\mathbf{b} \in \mathbb{R}^m$ is in the range of T is an **existence** question. Indeed, if $\mathbf{b} \in \text{Range}(T)$ then there **exists** a $\mathbf{x} \in \mathbb{R}^n$ such that $T(\mathbf{x}) = \mathbf{b}$. We now want to look at the problem of whether $\mathbf{x}$ is **unique**. That is, does there exist a distinct $\mathbf{y}$ such that $T(\mathbf{y}) = \mathbf{b}$.

Definition 8.9: A vector mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **one-to-one** if for each $\mathbf{b} \in \text{Range}(T)$ there exists only one $\mathbf{x} \in \mathbb{R}^n$ such that $T(\mathbf{x}) = \mathbf{b}$.

When T is a linear mapping, we have all the tools necessary to give a complete description of when T is one-to-one. To do this, we use the fact that if $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear then $T(\mathbf{0}) = \mathbf{0}$. Here is one proof: $T(\mathbf{0}) = T(\mathbf{x} - \mathbf{x}) = T(\mathbf{x}) - T(\mathbf{x}) = \mathbf{0}$.

Theorem 8.10: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Then T is one-to-one if and only if $T(\mathbf{x}) = \mathbf{0}$ implies that $\mathbf{x} = \mathbf{0}$.

If $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a matrix mapping then according to Theorem 8.10, T_A is one-to-one if and only if the only solution to $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$. We gather these facts in the following theorem.

Theorem 8.11: Let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a matrix mapping, where $A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n] \in M_{m \times n}$. The following statements are equivalent:

1. T_A is one-to-one.
2. The rank of A is $r = \text{rank}(A) = n$.
3. The columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent.

Example 8.12. Let $T_A : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be the matrix mapping with matrix

$$A = \begin{bmatrix} 3 & -2 & 6 & 4 \\ -1 & 0 & -2 & -1 \\ 2 & -2 & 0 & 2 \end{bmatrix}.$$

Is T_A one-to-one?

Solution. By Theorem 8.11, T_A is one-to-one if and only if the columns of A are linearly independent. The columns of A lie in $\mathbb{R}^3$ and there are $n = 4$ columns. From Lecture 6, we know then that the columns are not linearly independent. Therefore, T_A is not one-to-one. Alternatively, A will have rank at most $r = 3$ (why?). Therefore, the solution set to $A\mathbf{x} = \mathbf{0}$ will have at least one parameter, and thus there exists infinitely many solutions to $A\mathbf{x} = \mathbf{0}$. Intuitively, because $\mathbb{R}^4$ is “larger” than $\mathbb{R}^3$, the linear mapping T_A will have to project $\mathbb{R}^4$ onto $\mathbb{R}^3$ and thus infinitely many vectors in $\mathbb{R}^4$ will be mapped to the same vector in $\mathbb{R}^3$. $\square$

Example 8.13. Let $T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the matrix mapping with matrix

$$A = \begin{bmatrix} 1 & 0 \\ 3 & -1 \\ 2 & 0 \end{bmatrix}$$

Is T_A one-to-one?

Solution. By inspection, we see that the columns of A are linearly independent. Therefore, T_A is one-to-one. Alternatively, one can compute that

$$\text{rref}(A) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Therefore, $r = \text{rank}(A) = 2$, which is equal to the number columns of A . $\square$

8.3 Standard Matrix of a Linear Mapping

We have shown that all matrix mappings T_A are linear mappings. We now want to answer the reverse question: Are all linear mappings matrix mappings in disguise? If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear mapping, then to show that T is in fact a matrix mapping we must show that there is some matrix $A \in M_{m \times n}$ such that $T(\mathbf{x}) = A\mathbf{x}$. To that end, introduce the **standard unit vectors** $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ in $\mathbb{R}^n$:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

Every $\mathbf{x} \in \mathbb{R}^n$ is in $\text{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ because:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \dots + x_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots + x_n \mathbf{e}_n$$

With this notation we prove the following.

Theorem 8.14: Every linear mapping is a matrix mapping.

Proof. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear mapping. Let

$$\mathbf{v}_1 = T(\mathbf{e}_1), \mathbf{v}_2 = T(\mathbf{e}_2), \dots, \mathbf{v}_n = T(\mathbf{e}_n).$$

The co-domain of T is $\mathbb{R}^m$, and thus $\mathbf{v}_i \in \mathbb{R}^m$. Now, for arbitrary $\mathbf{x} \in \mathbb{R}^n$ we can write

$$\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots + x_n \mathbf{e}_n.$$

Then by linearity of T , we have

$$\begin{aligned} T(\mathbf{x}) &= T(x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots + x_n \mathbf{e}_n) \\ &= x_1 T(\mathbf{e}_1) + x_2 T(\mathbf{e}_2) + \dots + x_n T(\mathbf{e}_n) \\ &= x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 + \dots + x_n \mathbf{v}_n \\ &= [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_n] \mathbf{x}. \end{aligned}$$

Define the matrix $A \in M_{m \times n}$ by $A = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_n]$. Then our computation above shows that

$$T(\mathbf{x}) = x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 + \dots + x_n \mathbf{v}_n = A\mathbf{x}.$$

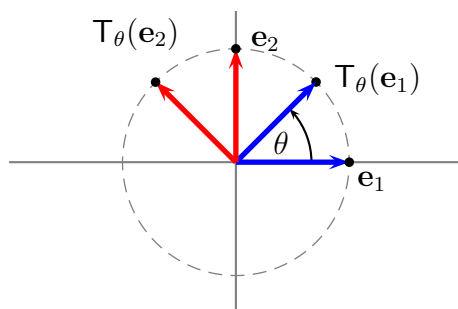
Therefore, T is a matrix mapping with the matrix $A \in M_{m \times n}$. □

If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear mapping, the matrix

$$\mathbf{A} = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n)]$$

is called the **standard matrix of T** . In words, the columns of $\mathbf{A}$ are the images of the standard unit vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ under T . The punchline is that if T is a linear mapping, then to derive properties of T we need only know the standard matrix $\mathbf{A}$ corresponding to T .

Example 8.15. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear mapping that rotates every vector by an angle θ . Use the standard unit vectors $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ in $\mathbb{R}^2$ to write down the matrix $\mathbf{A} \in \mathbb{R}^{2 \times 2}$ corresponding to T .



Solution. We have

$$\mathbf{A} = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2)] = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

□

Example 8.16. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a dilation of factor $k = 2$. Find the standard matrix $\mathbf{A}$ of T .

Solution. The mapping is $T(\mathbf{x}) = 2\mathbf{x}$. Then

$$T(\mathbf{e}_1) = 2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = 2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_3) = 2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

Therefore,

$$\mathbf{A} = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad T(\mathbf{e}_3)] = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

is the standard matrix of T .

□

After this lecture you should know the following:

- the relationship between the range of a matrix mapping $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ and the span of the columns of $\mathbf{A}$
- what it means for a mapping to be onto and one-to-one
- how to verify if a linear mapping is onto and one-to-one
- that all linear mappings are matrix mappings
- what the standard unit vectors are
- how to compute the standard matrix of a linear mapping